

Problem 1

Complete (4.B) from Homework 2.

The reactor in (4.B) is the same as in (4.A) from those solutions we can fill in the complete stoichiometric table as follows:

Sp.	ν_i	n_{i0} (mol·s ⁻¹)	n_i (mol·s ⁻¹)
A	-1	19.4	17.8
B	-1	174.8	173.1
C	1	0.0	1.7

The reactor is isothermal so the energy balance, Eq. (4.2), simplifies quite a bit: isothermal means $T = T_0$ so the first summation term is zero. With only one reaction the second sum consists of only a single term which, yielding

$$Q = Vr\Delta H$$

The previous work on (4.C) provided

$$\begin{aligned}\Delta H &= -116400 + 27(T - 298) \\ &= -116400 + 27(323 - 298) \\ &= -115\,725 \text{ J}\cdot\text{mol}^{-1}\end{aligned}$$

where the temperature is known because the system is isothermal. All that remains for this problem is to tackle the heat term, Q , given by Eq. (4.26) as

$$Q = UA(T_u - T).$$

The reaction is exothermic so in order to maintain an isothermal condition the utility will have to remove energy, and therefore its temperature will need to be lower than the reactor temperature, 50 °C. From our two options, water or steam, the water is the only viable option since its temperature is 15 °C.

It remains to determine U from Eq. (4.27). The utility is water so from Tab. 4.1 $h_u = 1500 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$, and the process stream is 90 mol% water so we assume its individual heat transfer coefficient is also nearly that of water, $h = 1500 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$. Therefore

$$U = \left[\frac{1}{h_u} + \frac{1}{h} \right]^{-1} = \left[\frac{1}{1500} + \frac{1}{1500} \right]^{-1} = 750 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}.$$

Substituting (4.27) into the energy balance above yields

$$\begin{aligned}(750 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}) A (15 - 50) &= (1.7 \text{ mol}\cdot\text{s}^{-1}) (-115\,725 \text{ J}\cdot\text{mol}^{-1}) \\ A &= \boxed{7.5 \text{ m}^2}\end{aligned}$$

In (4.A) we found $rV = 1.7 \text{ mol}\cdot\text{s}^{-1}$.

$T = T_0$ for isothermal CSTRs so

$$\sum n_{i0} C_{P,i} (T - T_0) = 0$$

Recall from (4.A) we found

$n_C = rV = 1.7 \text{ mol}\cdot\text{s}^{-1}$. Also, it's ok to take the temperature difference in °C since a degree change in °C is equal to a degree change in K.

Problem 2

6 mol·min⁻¹ pure A is fed to CSTR where the liquid phase, elementary reaction $A \longrightarrow 2B$ takes place with $k = 4 \text{ min}^{-1}$.

(2.A) Find X_A if $\tau = 10 \text{ s}$.

(2.B) Find n_A if second reactor added in series and with rate constant therein reduced by half.

(2.C) Find n_A if second reactor is instead added in parallel, inlet equally divided.

(2.D) Find n_B for both options, indicate which is better.

Part A

This appears to be a “minimum information” style problem and since conversion is mentioned we’ll have a good chance of making progress if we start with the conversion-based material balance for A given by Eq. (5.8) as

$$V = \frac{n_{A0}X_A}{-r_A}$$

Substituting the rate expression yields

$$V = \frac{n_{A0}X_A}{kC_{A0}(1-X_A)} = \frac{v_0}{k} \frac{X_A}{1-X_A}$$

which can be further rearranged to

$$X_A = \frac{k\tau}{1+k\tau}. \quad (2.A)$$

This will be the central equation we use for Parts B and C. Specifically for Part A we evaluate the conversion by substituting the given information $k = 4 \text{ mol} \cdot \text{min}^{-1}$ and $\tau = 10 \text{ s} = 10/60 = 1/6 \text{ min}$ to obtain

$$X_A = \frac{4/6}{1+4/6} = \boxed{0.4}$$

Part B

The second reactor follows Equation (2.A) but with a rate constant reduced by half. Denoting the series reactor with the superscript s the conversion of A in the series reactor, X_A^s , is

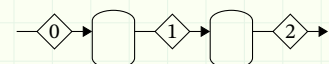
$$X_A^s = \frac{k^s\tau}{1+k^s\tau} = \frac{2/6}{1+2/6} = 0.25.$$

Labeling the streams as in the side bar the outlet molar flow rate of A is

$$\begin{aligned} n_{A2} &= n_{A1}(1-X_A^s) \\ &= n_{A0}(1-X_A)(1-X_A^s) \\ &= 6(1-0.4)(1-0.25) \\ &= \boxed{2.7 \text{ mol} \cdot \text{min}^{-1}} \end{aligned}$$

$$-r_A = kC_A = kC_{A0}(1-X_A)$$

$$k^s = 2 \text{ min}^{-1}$$



Part C

For the parallel system the flows into either reactor are reduced by half and therefore the residence time is doubled, so $\tau = 2/6$ min in Equation (2.A). The rate constant k is unchanged for the original reactor but is again halved for the second reactor. Using the notations shown in the side bar the outlet moles of A from the first reactor are

$$\begin{aligned} n_{A2} &= n_{A1} (1 - X_A) \\ &= \frac{n_{A0}}{2} \left(1 - \frac{k(2\tau)}{1 + k(2\tau)} \right) \\ &= 3 \left(1 - \frac{4(2/6)}{1 + 4(2/6)} \right) \\ &= 3(1 - 0.571) \\ &= 1.29 \text{ mol} \cdot \text{min}^{-1} \end{aligned}$$

and the outlet moles of A from the second reactor are

$$\begin{aligned} n_{A4} &= n_{A3} (1 - X_A) \\ &= \frac{n_{A0}}{2} \left(1 - \frac{(k/2)(2\tau)}{1 + (k/2)(2\tau)} \right) \\ &= 3 \left(1 - \frac{k\tau}{1 + k\tau} \right) \\ &= 3(1 - 0.4) \\ &= 1.80 \text{ mol} \cdot \text{min}^{-1} \end{aligned}$$

Therefore

$$n_{A5} = n_{A2} + n_{A4} = 1.29 + 1.80 = \boxed{3.09 \text{ mol} \cdot \text{min}^{-1}}$$

Part D

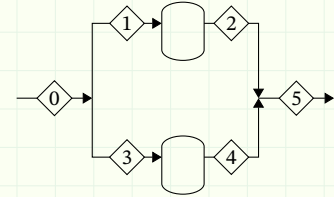
From a material balance on A the extent of reaction is $\xi = n_{A0} - n_A$ where n_{A0} refers to the *overall* moles entering a process and n_A refers to the *overall* moles exiting a process. On a similar basis the outlet moles of B are $n_C = 2\xi = 2(n_{A0} - n_A)$. Therefore for the series arrangement the outlet moles of B are

$$n_B = 2(6 - 2.7) = \boxed{6.60 \text{ mol} \cdot \text{min}^{-1}}$$

and for the parallel arrangement the outlet moles of B are

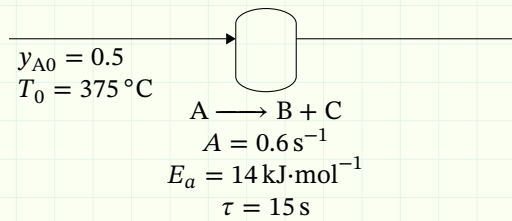
$$n_B = 2(6 - 3.09) = \boxed{5.83 \text{ mol} \cdot \text{min}^{-1}}$$

With a higher production rate of B the series arrangement is preferred.



Problem 3

Find the conversion of A (propane) for the elementary reaction shown below.



This is a “minimum information” style problem as evidenced by there being remarkably little information to go on. Therefore our stoichiometric table doesn’t tell us much of use, but it can still be helpful to write out:

Sp.	ν_i
A	-1
B	1
C	1
$\delta = 1$	

Since conversion is requested and there’s only one reaction, the conversion-based material balance of Eq. (5.8) is likely a good starting point. That’ll require the rate expression, which for this elementary, **gas-phase** reaction is

$$\delta = 1, \nu_A = -1, y_{A0} = 0.5.$$

$$\begin{aligned}
 -r_A &= r = kC_A \\
 &= kC_{A0} \frac{1 - X_A}{\left(1 - y_{A0} \frac{\delta}{\nu_A}\right) \frac{P_0}{P} \frac{T}{T_0}} \\
 &= kC_{A0} \frac{1 - X_A}{1 - (0.5) \frac{1}{-1} X_A} \\
 &= kC_{A0} \frac{1 - X_A}{1 + 0.5X_A}
 \end{aligned}$$

See Eq. (5.14) for C_A for gases.

Substituting in Eq. (5.8) yields

$$V = \frac{n_{A0}}{kC_{A0}} \frac{X_A (1 + 0.5X_A)}{1 - X_A}$$

which simplifies to

$$k\tau = \frac{X_A (1 + 0.5X_A)}{1 - X_A}.$$

We can find k since we’re given A and E_a , and τ is given from the problem statement as 15 s. Therefore this is one equation with one unknown. It could be solved with the quadratic equation but I’m lazy so I’ll plot it, look for a root, and use that as a guess for fsolve.

```

1 fplot(@rff, [0 1]); ylim([-3 3]);
2 % Plot suggests root around Xa = 0.4
3 XaSol = fsolve(@rff, 0.4)
4
5 function out = rff(Xa)
6     tau = 15; % s
7     T = 375+273; % K
8     A = 0.6; % 1/s

```

```
9   Ea = 14e3; % J/mol
10  R = 8.314; % J/mol-K
11  k = A*exp(-Ea/(R*T)); % 1/s
12  out = Xa.*(1+0.5.*Xa)./(1-Xa)-tau*k;
13 end
```

This code yields $X_A = 0.36$.

Problem 4

Liquid phase reaction is generically $2A + 4B \longrightarrow C + D + 4E$ with $r_A = -kC_A C_B$.

(Part A) Find expression for X_A . Plot from 0°C to 100°C .

(Part B) Find expression for $G(T)$.

(Part C) Plot $G(T)$ and $R(T)$.

(Part D) Estimate ignition temperature.

Additional information:

$$A = 2 \times 10^{10} \text{ L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$$

$$E_a = 70 \text{ kJ} \cdot \text{mol}^{-1}$$

$$C_{A0} = C_{B0} = 0.8 \text{ mol} \cdot \text{L}^{-1}$$

$$V = 0.5 \text{ L}$$

$$\tau = 50 \text{ s}$$

$$\Delta H = -1172 \text{ kJ} \cdot \text{mol}^{-1} \text{ (const.)}$$

Part A

Beginning with Equation 5.8,

$$V = \frac{n_{A0} X_A}{-r_A}$$

The net rate for A is

$$-r_A = kC_A C_B = kC_{A0}^2 (1 - X_A)(1 - 2X_A)$$

Substitution of $-A$ into Equation 5.8 yields

$$V = \frac{n_{A0} X_A}{kC_{A0}^2 (1 - X_A)(1 - 2X_A)}$$

which can be rearranged a little bit to

$$\begin{aligned} 1 &= \frac{n_{A0}}{kVC_{A0}^2} \frac{X_A}{(1 - X_A)(1 - 2X_A)} \\ &= \frac{1}{\beta} \frac{X_A}{(1 - X_A)(1 - 2X_A)} \end{aligned}$$

where $\beta = C_{A0}k\tau$ was defined to make rearrangement to the quadratic form a little easier to read and write. Expanding the denominator and moving it to the left side yields

$$\beta [1 - X_A + 2X_A^2 - 2X_A] = X_A$$

which can be rearranged to the standard quadratic form of Equation A.1 as

$$0 = (2\beta)X_A^2 + (-3\beta - 1)X_A + \beta.$$

From Equation 5.11,

$$C_A = C_{A0}(1 - X_A)$$

$$C_B = C_{A0}(1 - 2X_A)$$

$$\begin{aligned} \frac{n_{A0}}{kVC_{A0}^2} &= \frac{v_0}{kVC_{A0}} \\ &= \frac{1}{C_{A0}k\tau} \\ &= \frac{1}{\beta} \end{aligned}$$

For Equation A.1,

$$a = 2\beta$$

$$b = (-3\beta - 1)$$

$$c = \beta$$

Applying the quadratic equation yields

$$X_A = \frac{1 + 3\beta \pm \sqrt{(1 + 3\beta)^2 - 8\beta^2}}{4\beta}$$

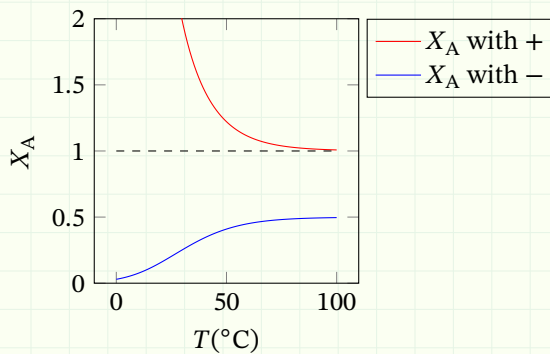
$$= \frac{1 + 3\beta \pm \sqrt{\beta^2 + 6\beta + 1}}{4\beta}$$

which is the form requested with $\beta = C_{A0}k\tau$. Plotting these two expressions yields the following:

```

1 Ca0 = 0.8; % mol/L
2 tau = 50; % s
3 k = @(T) 2e10*exp(-70000./(8.314*T)); % L/mol-s
4 B = @(T) Ca0*k(T)*tau; % non-dim
5 XaPlus = @(T) (1+3*B(T) + sqrt(B(T).^2+6*B(T)+1))./(4*B(T));
6 XaMinus = @(T) (1+3*B(T) - sqrt(B(T).^2+6*B(T)+1))./(4*B(T));
7 hold on
8 fplot(XaPlus, [0,100]+273, 'r')
9 fplot(XaMinus, [0,100]+273, 'b')
10 yline(1, 'k--')
11 hold off
12 axis square; grid on; box on;
13 xlabel('T (K)'); ylabel('X_A');
14 legend('+sqrt', '-sqrt')
15 ylim([0 2])

```



Only the term with $-\sqrt{\dots}$ is physically reasonable because X_A must be less than 1, so

$$X_A = \frac{1 + 3\beta - \sqrt{\beta^2 + 6\beta + 1}}{4\beta}$$

Part B

From Equation 5.16,

$$G(T) = -Vr\Delta H$$

$$= -\frac{1}{2}kVC_A C_B \Delta H$$

$$= -\frac{1}{2}kVC_{A0}^2 \Delta H (1 - X_A)(1 - 2X_A)$$

$C_{A0} = 0.8 \text{ mol}\cdot\text{L}^{-1}$ and $\tau = 50 \text{ s}$, both of which are not functions of temperature. The rate constant k must be calculated as

$$k = (2 \times 10^{10}) \exp\left[\frac{-70000}{8.314T}\right]$$

for each value of T from 0°C to 100°C .

r_A is related to r as

$$r_A = -2r \quad \Rightarrow \quad r = -\frac{1}{2}r_A$$

Substituting for r_A yields

$$r = \frac{1}{2}kC_A C_B$$

Part C

From Equation 5.16,

$$R(T) = \sum n_{i0} C_{P,i} (T - T_0) \approx n_{T0} C_P (T - T_0)$$

where the approximation made is that the inlet is pure water. The total inlet molar flow rate is therefore

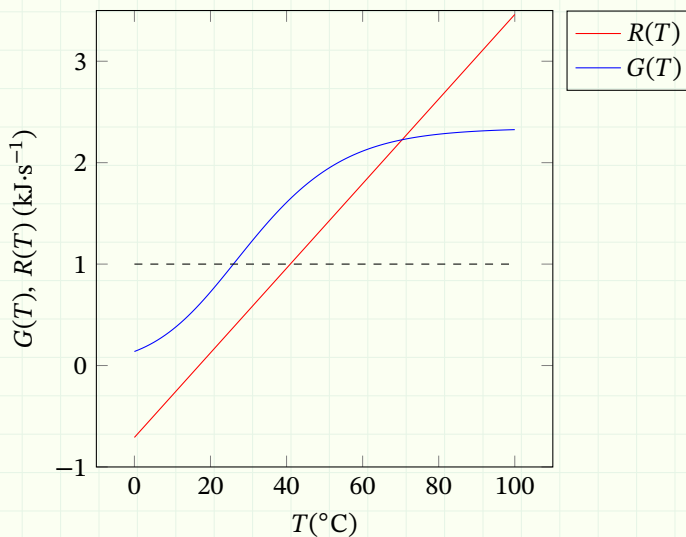
$$n_{T0} = \frac{\rho v_0}{MM} = \frac{\rho V}{\tau MM} = \frac{(1)(0.5)}{(50)(0.018)} = 0.556 \text{ mol}\cdot\text{s}^{-1}$$

from which $R(T)$ is approximated as

$$R(T) \approx 41.7(T - 290)$$

Assuming the script from Part A has been run then the plot of $G(T)$ and $R(T)$ can be made as follows:

```
1 % Run script for Part A before this script
2 Xa = @T) XaMinus(T); % from Part A
3 dH = -1172e3; % J/mol
4 V = 0.5; % L
5 G = @T) -0.5*V*(Ca0^2)*dH*k(T).*(1-Xa(T)).*(1-2.*Xa(T))/1000;
   % kJ/s
6 R = @T) 41.7*(T-290)/1000; % kJ/s
7 hold on
8 fplot(G, [0,100]+273, 'r')
9 fplot(R, [0,100]+273, 'b')
10 hold off
11 axis square; grid on; box on;
12 xlabel('T (K)'); ylabel('G(T) or R(T) (kJ/s)');
13 legend('G(T)', 'R(T)')
```

**Part D**

From the left plot in Figure 5.5 the extinction temperature is the value of T_0 which produces the upper metastable point. From the fourth row in Figure 5.4 the upper metastable state occurs when $R(T)$ is tangent to the high-temperature inflection point in $G(T)$. Therefore we have to vary T_0 until $R(T)$ is (approximately) tangent to $G(T)$ with $R(T)$ to the left of $G(T)$. Whatever value of T_0 accomplishes this is the extinction temperature. The difference in

For water,

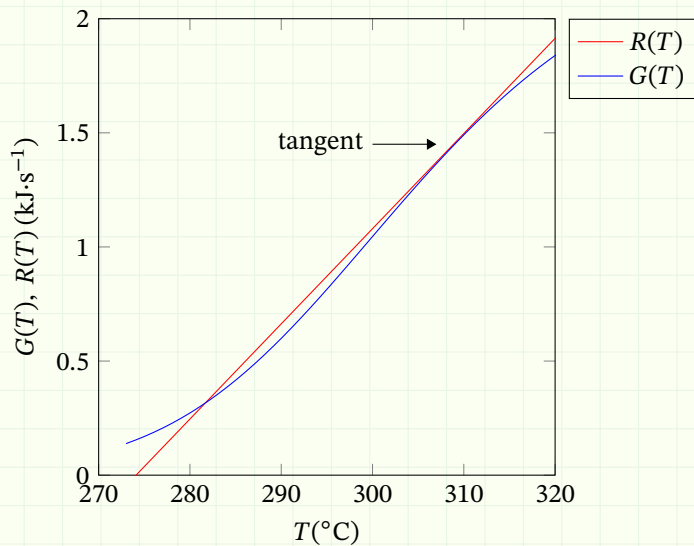
$$\rho = 1 \text{ kg}\cdot\text{L}^{-1}$$

$$MM = 0.018 \text{ kg}\cdot\text{mol}^{-1}$$

$$C_P = 75 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$$

the two reactor temperatures for the two steady states in this condition is the change in reactor temperature when extinction occurs and we can evaluate X_A at these two temperatures to calculate the drop in temperature.

Setting T_0 to about 274.1 K (about 0.9°C) (which is the extinction temperature) produces



from which the reactor temperature change is approximately 310 K to 280 K, or from about 37°C to 7°C . The corresponding change in conversion is obtained by evaluating X_A at these two temperatures which produces a change from 32% to 6% conversion.